From Bias to Brilliance: The Impact of Artificial Intelligence Usage on Recruitment Biases in China

Fei Zheng, Chenguang Zhao, Muhammad Usman, and Petra Poulova

***Abstract*—In the rapidly evolving landscape of human resources and talent acquisition, the impact of the usage of artificial intelligence (hereafter, AI) on recruitment biases has emerged as a pivotal and transformative subject of study. Therefore, this study aims to critically evaluate the impact of AI usage on recruitment biases, particularly in the context of China. The data were gathered through a survey of 423 respondents working in the manufacturing sector. We use a cross-sectional dataset and various diagnostics (i.e., reliability and collinearity tests). The empirical findings using multivariate regression techniques suggested that Al usage is reshaping the recruitment process by offering innovative solutions to tackle biases that have pervaded the hiring process for years. However, human involvement is indispensable in the recruitment process, alongside the use of AI. Although the use of AI can efficiently handle tasks such as resume screening and data analysis, human judgment brings essential qualities to the hiring process. Human recruiters possess the ability to assess a candidate's soft skills, cultural fit, and emotional intelligence, as these qualities are challenging for AI to comprehend. The policy implications of the study recommend that by combining the strengths of AI efficiency with human insight, organizations can create a recruitment process that is not only objective and efficient but also considerate, ethical, and aligned with the values and goals of the company.**

***Index Terms*—Artificial intelligence (AI), human resource management, recruitment biases.**

I. Introduction

THE world is currently more interconnected than ever. As a way to compete in the worldwide labor market, corporations are undergoing transformational change in anticipation of swiftly moving developments in technology [1], [2]. In addition to the concentration of job responsibilities during the internationalization era, there is significant rivalry for the few, superior staff members necessary for businesses to remain competitive internationally and satisfy ever-rising demands from consumers [3]. Proper human resource planning is essential since hiring the wrong people or refusing to accept changes to the traditional recruiting process can be costly [4]. Unfortunately, global talent shortages have made it more challenging than ever to find, hire, and keep excellent employees [5].

The increasing significance assigned to human resources is one of the most fundamental recent advances in the field of business. People are essential for enterprises because they bring various perspectives, concepts, and character traits to the workplace [6]. Furthermore, if handled professionally, these characteristics could be of immense benefit to the company as a whole [7]. The process of identifying and recruiting prospective employees from both inside and outside of a corporation to start analyzing them for future job opportunities begins with recruitment, while selection starts after the right individuals are identified [8]. However, the corporation gets the chance to project a favorable image through the hiring and termination of staff members [9]. Furthermore, it has been suggested that a company needs effective employees to build and maintain a competitive advantage [10], [11].

Organizations now need to hire personnel quickly, in sufficient numbers, and with the right qualifications, making recruitment and selection crucial [12], [13]. Companies typically directly correlate the effectiveness of their businesses with the people they hire [14], [15]. Therefore, it is essential to hire the right people to achieve organizational success [10], [16]. Managers are becoming more aware that using data analytics to analyze and manage human resources and boost employee and organizational performance may be more advantageous than relying solely on intuition and experience [17], [18].

Artificial intelligence (AI) is vital to humanity, society, and business. Most companies are changing their strategies and business structures to incorporate AI [19], [20], [21]. AI is advancing technology and business [22], [23]. Large datasets and faster processing power boost AI [24], [25]. AI can automate and optimize routine processes to save time and money, increase productivity, use cognitive-based technologies to make business decisions, avoid human error, predict user preferences, create high-quality leads, raise profits, and provide intelligent guidance and assistance [26], [27]. Managers realize that using data analytics to assess and manage human resources, as well as improve employee and organizational performance, maybe better than intuition and experience [28]. Human-like robots employ AI to mimic human intelligence [29], [30]. These robots may see, hear, decide, and translate languages like humans [31], [32]. Most of

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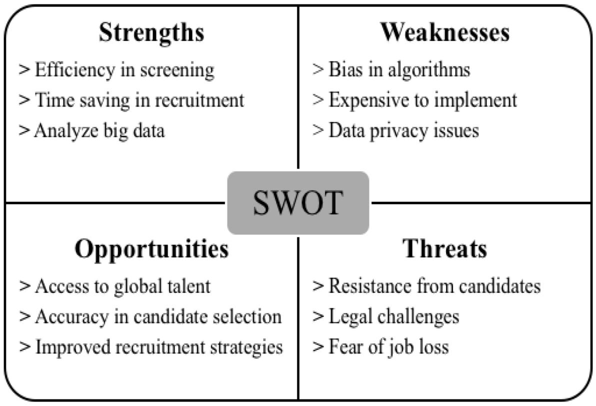

Fig. 1. SWOT matrix.

these tools used natural language processing (NLP), machine learning (ML), and algorithm analysis (AA) [33].

Multiple human resource recruitment biases exist that can affect the hiring process [34], [35], [36]. These can occur in several ways, one of which is when human recruiters unintentionally favor applicants based on characteristics like gender, race, ethnicity, disability, or social background [37], [38]. Companies may use AI to create more unbiased and equitable hiring processes [39]. This article aims to investigate how AI may support companies in developing a more equal, inclusive, and efficient hiring process that may eventually lead to better results for both the company and its staff. Recent debates [25], [29], [38] have opened up new avenues for studying how recruiters' biased opinions and judgments can influence the recruitment process, and how modern tools like AI can significantly improve the hiring process. Therefore, the core objective of this study is to address how the AI usage can guide the overall recruitment process and make it unbiased. Moreover, we also draw a SWOT matrix, which explains the pros and cons of AI technology in personnel recruitment (see Fig. 1).

The rest of this article is organized as follows. Section II provides the most recent literature on the topic and discusses the different recruitment biases and how the use of AI tools could minimize these biases. In Section III, the methodology is being elaborated. In Section IV, an empirical analysis has been conducted. In Section V, the discussion has been made regarding the empirical findings. Finally, Section VI concludes the topic and provides implications, limitations, and future research directions.

II. Theoretical Background

A. Review of Literature

The dynamic landscape of recruitment biases and AI continues to evolve, and recent literature reflects the ongoing quest to understand and address this critical issue. In the face of rapid technological advancements, the most current literature emphasizes the need to scrutinize AI-driven recruitment systems for emerging biases and explore innovative strategies to mitigate them. Therefore, Table I provides a summary of the literature. This review aims to capture the state of the art by examining the latest research, methodologies, and insights that shape our understanding of how AI is reshaping recruitment practices and its implications for diversity, equity, and fairness in hiring. Furthermore, it helps identify a gap in the existing literature.

The literature review on the impact of AI on recruitment biases reveals several significant gaps in existing research. For example, most of the studies either observed the role of AI in strengthening the recruitment process [45], [46], [48], [49] or utilized AI from the recruiter's perspective [41], [43], [47]. Furthermore, while studies by Horodyski [40] and Uma et al. [44] showed that using AI recruitment methods can help reduce biases in the traditional hiring process, there is still a big knowledge gap when it comes to fully understanding the specific biases that AI faces and how they affect recruitment outcomes, especially in China.

As disruptive technologies continue to reshape industries and bring about corporate change, AI integration in recruitment processes has become increasingly common [20]. However, while AI promises efficiency and objectivity, it also brings the cost of a tradeoff between humans and machines. By relying on historical data, AI systems may inadvertently replicate patterns of discrimination present in past hiring decisions. Businesses must actively work to mitigate these biases by continually refining AI algorithms, ensuring diverse training data, and implementing robust oversight mechanisms. Failure to address these biases not only undermines the potential of AI in recruitment but also perpetuates systemic inequalities within the workforce.

B. Hypotheses Development

Recruitment biases in human resources are unintentional or intentional decisions that can affect the hiring process and lead to the unfair treatment of some applicants based on factors including gender, race, age, ethnicity, ability, or cultural standing [34]. The effects of AI in managing human resources, particularly the recruitment and selection process, need to be considered in light of the advancement in technology [50]. We particularly focus on China for several reasons. First, the Chinese labor market is one of the world's largest and most diverse [51]. This diversity provides solid ground to study how AI technologies could impact the recruiting process in various circumstances. Studying Chinese recruitment practices can help researchers learn about the problems and limitations of adopting AI tools in a complex job market. Second, China is leading in AI research and implementation because it has an ambitious national plan that considers AI as an element of growth [52]. This rapid adoption of AI in many areas of life, including recruitment, is a dynamic case study of how technology changes hiring. In this study, we are not going to explore the causal relationship, i.e., the reasons for recruitment biases or their impact on different outcomes. However, in the following paragraphs, we highlight the recruitment biases individually and then explain how the use of AI can overcome them.

1) Conformity Bias: Conformity bias occurs when a recruiter focuses on information that supports their preconceived notions about a candidate while ignoring information that contradicts those notions [53]. For example, even if a female candidate

TABLE I
LITERATURE REVIEW

Author – Year	Key Findings
Horodyski [40]	The study's findings indicate that applicants generally view AI technology positively in recruitment processes, finding it valuable and user-friendly. Notably, they recognize the most significant advantage as a substantial reduction in response time. However, applicants also identify key drawbacks, such as AI's inability to replicate nuanced human judgment, accuracy and reliability issues, and a perception of AI in recruitment as being in its early stages of development.
Horodyski [41]	AI technologies have reached a stage where they can automate substantial portions of the hiring process, consequently reshaping the roles of recruiters and HR professionals. Nonetheless, there's limited research on how recruiters perceive and adopt AI, and the factors influencing its utilization remain relatively unexplored. This study, using a web-based survey with 238 participants representing diverse demographics, examined the intentions of recruiters to incorporate AI into their workflow. By extending the Unified Theory of Acceptance and Use of Technology (UTAUT) to consider AI use frequency and education, the study uncovered significant findings.
Varsha [42]	Coded algorithms play a pivotal role in decision-making within firms, yet they can harbor various biases and uncertainties. This qualitative study underscores that the biases and vulnerabilities encountered by AI across industries have significant repercussions, leading to gender and racial biases. The study comprehensively categorizes these biases and underscores the paramount importance of responsible AI implementation in organizations to mitigate AI-related risks.
França, et al. [43]	This study offers valuable insights into the intersection of AI and Human Resource Management. To ensure the reliability of the findings, the authors conducted a systematic literature review following the PRISMA guidelines, effectively minimizing bias. The discussions center on critical themes such as AI's impact on talent management, AI bias, ethics and legal considerations, and their
	implications for HR management. Importantly, the findings stress the proactive role HR managers must play in adopting technology and bridging gaps—be they technological, human, societal, or governmental.
Uma, et al. [44]	Findings from the scoping review of the literature reveal that the incorporation of automation and analytics into the recruitment and selection processes has a significant impact. These technology-driven approaches effectively eliminate biases that are increasingly prevalent in traditional manual hiring practices. Additionally, research suggests that candidates often employ impression management tactics in traditional face-to-face interviews, but these tactics can be mitigated through automated recruitment methods.
Gupta and Mishra [45]	The examination of the current landscape reveals that while many companies have begun to incorporate AI tools into their recruitment processes, there remains untapped potential in leveraging a broader spectrum of algorithms for the entire recruitment and selection journey. This widespread adoption of AI presents both opportunities and challenges for practitioners in the realm of Human Resource Management.
Sharma, et al. [46]	The analysis indicates a steady rise in the adoption of AI technologies across HR practices. AI's role extends to various facets of HR, encompassing recruitment, selection, training, development, and resume scanning, among others. The integration of AI into HRM offers numerous advantages to organizations, including enhanced employee engagement, improved relationships, heightened competitive advantages, and optimized utilization of HR budgets. AI in HRM and delves into strategies to address these challenges, shedding light on AI's evolving role within HR practices through a comprehensive review of case studies, literature studies, and analyses of leading global organizations' AI applications in HRM.
Veglianti, et al. [47]	Recent years have witnessed a rapid expansion of technological innovations in e-recruitment systems, offering Human Resource professionals advanced tools to identify the most suitable talent for their organizations. This paper's primary objective is to investigate how AI Technologies can enhance recruitment efficiency and mitigate human errors by examining the theoretical alignments across diverse approaches and platforms designed for corporate use. The study employs a case study approach to address the research inquiries.
Aamer, et al. [48]	AI plays a pivotal role in this process, with its ultimate aim being the automation of tasks traditionally performed by humans. Remarkably swift and precise, AI operates and reacts in a manner akin to human capabilities. This study delves into the impact of AI on the Human Resource industry, with a specific focus on its application in the recruitment and selection process. The findings underscore that AI technology holds the potential to substantially enhance Human Resource and recruitment operations, offering benefits such as heightened productivity, reduced costs, enhanced accuracy, decreased workload, and an improved candidate experience.
Palos-Sánchez, et al. [49]	In organizations, the prevalence of AI is on the rise, and within the realm of Human Resource Management, AI's relevance has seen a notable surge in recent years. This article conducts a bibliometric analysis of scientific literature that systematically explores the utilization and influence of AI in HRM. The findings underscore that the application of AI in HRM is a burgeoning field marked by continuous expansion and optimistic prospects. However, it is noteworthy that the research has a distinct focus on AI's role in recruitment and selection processes, leaving other promising sub-areas relatively underexplored.

has the required abilities and competence, a recruiter may think she is uninterested in a technical career. AI could reduce this prejudice in a variety of ways. We can teach AI algorithms using various informational databases that feature candidates from underrepresented groups. Regular assessments of AI systems can prevent the propagation of prejudices in hiring procedures, thereby mitigating the potential biases resulting from the use of data sets biased against specific demographics. This may entail examining hiring data to identify bias trends associated with hiring problems.

Hypothesis 1: AI usage may be negatively associated with conformity bias.

2) Affinity Bias: Affinity bias occurs when a recruiter favors candidates who share their gender, race, or background [54]. For example, even though the other candidates have more qualifications, the recruiters could feel more at ease with a candidate who went to the same university as them. To lessen prejudice, AI can use objective criteria to evaluate applicants, such as skill assessments or project standards. This can help remove biases that may arise from evaluating applications inferentially. Therefore, organizations may ensure that they choose the most competent and diverse people for their employment opportunities by using AI to reduce affinity biases in recruiting.

2) Hypothesis 2:: AI usage may be negatively associated with affinity bias.

3) Halo Bias: The halo bias happens when a recruiter notices one good quality in a candidate and assumes that all other qualities must also be good [55]. For example, a recruiter can assume, despite the lack of evidence, that a candidate with a strong resume must also have strong communication skills. Utilizing AI algorithms to review applicant data and generate standardized assessments that paint a more comprehensive and objective picture of a candidate's skills, experience, and potential is one method for decreasing prejudice. AI can assist with the preparation and conduct of structured interviews that use predetermined questions to evaluate applicants. AI can look at historical hiring data to find patterns and trends that could improve decision-making.

Hypothesis 3: AI usage may be negatively associated with halo bias.

4) Recency Bias: Recency bias occurs when a manager bases the final choice of the application on the most recent event, such as the candidate's excellent interview, even though this does not guarantee his/her strong job performance once the applicant joins the company [56]. For example, during the shortlisting process, interviewers look out for candidates' preconceived responses. Interviewers select candidates for the post based on their responses to preconceived questions. We could lessen these biases by using various AI techniques. AI can scan audio, video, and text data from candidate interviews to identify trends and draw attention to issues or areas of interest. This method can help recruiters eliminate recency bias by providing objective data to evaluate candidates' responses rather than solely relying on their memories and impressions. AI can align individuals' talents and experience with job requirements, enabling recruiters to scrutinize candidates who might have overlooked relevant past experiences.

Hypothesis 4: AI usage may be negatively associated with recency bias.

5) Horn Bias: The horn bias occurs when a person's perception of another person is tainted by a single unfavorable aspect or characteristic [57]. For instance, we reject qualified and suitable individuals based on a single unfavorable factor, such as race or ethnicity. We can use AI algorithms to uniformly evaluate applicants according to a set of predetermined norms. AI can conceal sensitive personal information such as name, gender, age, and race from applicant profiles during the first screening phase of the hiring process, thereby minimizing the risk of subjective assessment and the negative effects of the horn effect. Age and race help to reduce the likelihood of subjective evaluation and the horn effect's negative effects. AI can incorporate feedback from a variety of sources, including managers, peers, and subordinates, to provide a more thorough assessment of a candidate's suitability for the position. By analyzing the vocabulary in job descriptions, hiring materials, and candidate profiles, AI can identify potentially biased language or tone that could contribute to the horn effect.

Hypothesis 5: AI usage may be negatively associated with horn bias.

6) Contrast Bias: Contrast bias occurs when there is a sizable difference between the outcomes of two or more candidates [58]. For example, the outcome for the second applicant will not be favorable if the first applicant exhibits exceptional performance

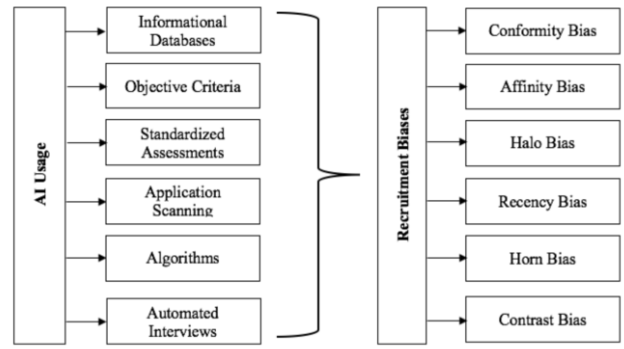

Fig. 2. Theoretical framework.

in the interview. Regardless of the second applicant's performance compared to the first, there is a possibility of rejection. Organizations can reduce contrast bias resulting from personal preferences or beliefs by using AI-based tools to evaluate candidates' competence. AI-powered chatbots and automated interviews can standardize the hiring process and ensure uniform questioning of all candidates. This technique assists in removing insider biases based on the opinions or preferences of interview subjects.

Hypothesis 6: AI usage may be negatively associated with contrast bias.

7) Overall Recruitment Biasness: Organizations can reduce the stated biases by using effective and efficient AI tools. We use machine learning algorithms to analyze resumes, job titles, and other hiring data, thereby reducing unconscious biases in the hiring process. AI can also analyze job descriptions and identify language that might be discriminatory or exclusionary, such as sexist terms or language that emphasizes characteristics unique to particular groups. AI has the potential to develop specialized pre-employment tests that assess a candidate's abilities and skills without depending on subjective evaluations.

Hypothesis 7: AI usage may be negatively associated with overall recruitment bias.

As illustrated in Fig. 2, the use of AI, particularly in the process of recruitment, could reduce the chances of various biases by incorporating algorithms and criteria that previously relied on humans. AI systems use informational databases that are free from prejudice, criteria that are not influenced by gender, origin, or other characteristics, objective tests, examination of applications and code, sophisticated algorithms, and data-based interviews to ensure an unbiased assessment of the candidates. This approach also minimizes the impact of various biases like conformity, affinity, halo, recency, horn, and contrast biases; therefore, the recruiters only consider the facts, objective performance, and criteria while hiring potential candidates.

III. Methodology

A. Data, Population, and Sample

The objective of this study is to investigate the effects of AI usage on biases in the recruitment process. To achieve the said objective, this study focuses on Chinese individuals working in the manufacturing sector. The selection of this sector is because of its prominence in China, which provides employment to the largest segment of the workforce. Targeting this sector, subsequently ensures that a significant portion of the working population has been captured. To gather the statistical data, we use a questionnaire-based technique. The questionnaire is constructed in Chinese, which is the official language and is widely understood by Chinese workers. This choice is crucial for ensuring that the participants fully understand the questions, which eventually leads to more accurate and meaningful responses. Moreover, the questionnaire is accompanied by a cover letter explaining the research objective. This letter serves multiple purposes. First, it ensures that participants are aware of the objectives of the study. Second, it motivates participants to provide thoughtful and honest responses.

The study also takes some steps to minimize potential preconceptions, because there are several numerical data issues regarding the quality [59]. For example, respondents are assured of complete confidentiality. Moreover, by informing respondents that there are no right or wrong answers, the study encourages them to answer truthfully. Based on the nature of the study, we use stratified random sampling techniques. Focusing on the manufacturing sector further narrows down the population, makes it easier to manage, and ensures reliability. Similarly, within the manufacturing sector, this study ensures diversity in terms of job roles, experience levels, and geographic locations, thus providing a comprehensive understanding of how the usage of AI tools can impact recruitment biases [60], [61]. A total of 650 questionnaires were distributed to collect data, out of which 423 responses were received; thus, the response rate is around 65%.

B. Variables Measurement

Table II explains the construction of dependent, independent, and control variables, along with their measuring scales and supportive literature. Our study specifically focuses on recruitment biases, categorizing them into seven distinct types. Similarly, the independent variable in this study is AI. As AI is a technology and cannot be directly measured using survey instruments, we use two proxies to quantify AI. The first proxy is the usage of AI tools, which demonstrates the use of AI in the manufacturing sector, particularly for the hiring of employees. The second proxy to measure AI is fairness in the usage of AI tools. The said proxy quantifies the effectiveness of using AI tools. Finally, we use corporate-level demographics as the control variables in this study.

C. Variables Measurement

By considering the objectives of the study, we draw a conceptual framework (see Fig. 3). The conceptual framework depicts

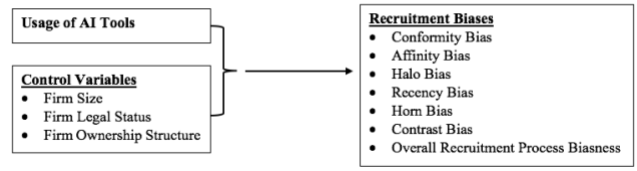

Fig. 3. Theoretical framework.

the relationship between independent and control variables, as well as the dependent variable, in a pictorially presented manner. It also aligns with the aforementioned hypotheses.

We use parametric analysis to figure out the empirical model and determine how much the use of AI affects recruitment biases. This means that we use certain statistical methods to see how independent variables and control variables affect the dependent variables. Survey datasets typically favor parametric analysis due to several advantages. First, parametric tests, such as multivariate regressions, are statistically significant, which makes them effective in identifying the impact. Second, parametric analysis provides interpretable results that have clear and intuitive interpretations. Third, parametric analysis is suitable for handling a variety of variables that are common in survey datasets. In particular, we have developed the empirical model below, which supports our empirical analysis and hypothesis testing

$$\begin{aligned}\text{Recruitment Biases}_i = \; & \beta_0 + \beta_1 \,\text{Usage of AI Tools}_i \\ & + \sum_{j\,=\,1}^{06} \beta_j \,\text{Control Variables}_i + \varepsilon_i.\end{aligned}$$

The aforementioned equation represents regression models where the dependent variable is $\text{Recruitment Biases}_\text{i}$ that are further segregated into conformity, affinity, halo, recency, horn, contrast, and overall recruitment biases. Moreover, $\text{Usage of AI Tools}_i$ represents an independent variable. Finally, $\text{Control Variables}_i$ are separated into firm size, legal status, and ownership structure. β_0 is an intercept term representing the expected value of $\text{Recruitment Biases}_i$ when all independent and control variables are zero. ε_i is an error term, representing the difference between the observed value of $\text{Recruitment Biases}_i$ and the value predicted by the model. It captures the effects of unobserved factors and random variation.

IV. Empirical Analysis

The sample of the study consists of 423 respondents from China ($N = 423$). Table III provides detailed descriptive statistics about the respondents. According to the respondents' provided information, 30% work in domestic-owned firms, 38% in foreign firms, and 32% in foreign and government-owned firms, respectively. According to their legal status, 40% of respondents work in sole proprietorship firms, 16% in partnership firms, and 44% in companies. Whereas 24% of respondents are working in small firms, 28% are in medium-sized firms, and 48% are in companies.

TABLE II
VARIABLE DEFINITIONS

Variables Name	Measurement	Scale	Supportive Literature
Dependent Variables			
Conformity Bias	The interviewers seemed to expect candidates to conform to a specific set of beliefs or behaviors.	5-Point Likert Scale – (1-5 Strongly Agree-Strongly Disagree) Higher value represents lesser biasness	Liu, et al. [53]
Affinity Bias	Interviewers showed a preference for candidates with similar backgrounds or interests as themselves.	5-Point Likert Scale – (1-5 Strongly Agree-Strongly Disagree) Higher value represents lesser biasness	Contreras, et al. [54]
Halo Bias	Positive attributes or qualifications of some candidates overshadowed other aspects of their candidacy.	5-Point Likert Scale – (1-5 Strongly Agree-Strongly Disagree) Higher value represents lesser biasness	Schmidt, et al. [55]
Recency Bias	The interview questions focused primarily on my most recent job or experience, neglecting my earlier achievements or experiences.	5-Point Likert Scale – (1-5 Strongly Agree-Strongly Disagree) Higher value represents lesser biasness	Cakici and Zaremba [56]
Horn Bias	A single mistake or shortcoming in my application or interview seemed to overshadow my other qualifications.	5-Point Likert Scale – (1-5 Strongly Agree-Strongly Disagree) Higher value represents lesser biasness	Rowley, et al. [57]
Contrast Bias	I felt that I was compared to other candidates during the interview process.	5-Point Likert Scale – (1-5 Strongly Agree-Strongly Disagree) Higher value represents lesser biasness	Salimi and Chandrasekar [58]
Recruitment Process Biasness	Overall, I felt the recruitment process was not fair and unbiased.	5-Point Likert Scale – (1-5 Strongly Agree-Strongly Disagree) Higher value represents lesser biasness	Thomas and Reimann [34]
Independent Variable			
Usage of AI Tools	AI tools, such as resume screening algorithms, were used in my recruitment process.	5-Point Likert Scale – (1-5 Strongly Disagree-Strongly Agree) Higher value represents higher usage of AI tools	Trivedi [31]
Fairness in the Usage of AI Tools	I believe AI tools were used fairly and without bias in my recruitment process.	5-Point Likert Scale – (1-5 Strongly Disagree-Strongly Agree) Higher value represents fairness in the usage of AI tools	
Control Variables			
Firm Size	Small	Dummy variable, value is "1" if firm has number of employees from 5 to 19, and "0" otherwise.	Wickstrøm and Bager [62]
	Medium	Dummy variable, value is "1" if firm has number of employees from 20 to 99, and "0" otherwise.	
	Large	Dummy variable, value is "1" if firm has number of employees more then 100, and "0" otherwise.	
Firm Legal Status	Sole Proprietorship	Dummy variable, value is "1" if firm is sole proprietorship, and "0" otherwise.	Chen [63]
	Partnership	Dummy variable, value is "1" if firm is partnership, and "0" otherwise.	
	Company	Dummy variable, value is "1" if firm is company, and "0" otherwise.	
Firm Ownership Structure	Domestic	Dummy variable, value is "1" if owned by domestic individual, companies or organizations, and "0" otherwise.	Malmendier, et al. [64]
	Government	Dummy variable, value is "1" if owned by government or state, and "0" otherwise.	
	Foreign	Dummy variable, value is "1" if owned by foreign individuals, companies or organizations, and "0" otherwise.	

Most variables have skewness values close to 0, indicating approximately symmetric distributions. A few variables have slightly right-skewed (positive skewness) or left-skewed (negative skewness) distributions, but the skewness values are relatively small. Many variables have negative kurtosis values, indicating platykurtic distributions (light tails). However, the kurtosis values are close to 0, suggesting approximately normal distributions for most variables. A few variables have moderately negative kurtosis values, indicating platykurtic distributions, and a few have moderately positive kurtosis values, indicating leptokurtic distributions (heavy tails).

Regarding the validity of empirical analysis, we have adopted both internal and external validation. To internally validate the findings of the study, we have incorporated the Cornbrash alpha technique. The said test is an important measure to check the reliability and validity of empirical data and show how close the set of items is to the group. It is also considered a factor to check the reliability of the scale. In empirical analysis, a high alpha value represents higher consistency and validity of the construct. It also specifies that the items used are suitable together on a cohesive scale. According to the findings of the said test reported in Table IV, the variables of this study are not only reliable

TABLE III
DESCRIPTIVE STATISTICS

Variables	Observation	Mean	Standard Deviation	Minimum	Maximum	Skewness	Kurtosis
Use of AI Tools	423	2.74	1.06	1	5	0.01	−0.47
Fairness in Use of AI Tools	423	2.88	0.97	1	5	−0.21	−0.06
Conformity Bias	423	2.12	0.62	1	5	−0.18	−0.35
Affinity Bias	423	2.28	0.92	1	5	0.25	−0.55
Halo Bias	423	2.4	0.78	1	5	0.06	−0.64
Recency Bias	423	2.44	1.04	1	5	0.01	−0.36
Horn Bias	423	2.4	0.9	1	4	−0.03	−0.53
Contrast Bias	423	2.54	1.15	1	5	0.18	−0.23
Recruitment Process Biasness	423	2.76	1.12	1	5	−0.08	−0.42
Domestic Owned Firms	423	0.3	0.47	0	1	0.6	−1.51
Foreign Owned Firms	423	0.38	0.47	0	1	0.23	−1.85
Government Owned Firms	423	0.32	0.47	0	1	0.47	−1.67
Sole Proprietorship Firms	423	0.4	0.49	0	1	−0.13	−1.49
Partnership Firms	423	0.16	0.37	0	1	1.20	0.46
Company	423	0.44	0.49	0	1	−0.05	−1.84
Small Size Firms	423	0.24	0.5	0	1	0.97	0.32
Medium Size Firms	423	0.28	0.44	0	1	0.78	−0.58
Large Size Firms	423	0.48	0.43	0	1	−0.48	−1.39

TABLE IV
RELIABILITY TEST

Items	Observations	Average Interitem Correlation	Alpha
Use of AI Tools	423	0.09	0.74
Fairness in Use of AI Tools	423	0.09	0.73
Conformity Bias	423	0.09	0.73
Affinity Bias	423	0.10	0.76
Halo Bias	423	0.09	0.72
Recency Bias	423	0.09	0.73
Horn Bias	423	0.12	0.79
Contrast Bias	423	0.11	0.77
Recruitment Process Biasness	423	0.10	0.76
Domestic Owned Firms	423	0.09	0.74
Foreign Owned Firms	423	0.11	0.78
Government Owned Firms	423	0.10	0.76
Sole Proprietorship Firms	423	0.09	0.74
Partnership Firms	423	0.11	0.77
Company	423	0.10	0.76
Small Size Firms	423	0.11	0.78
Medium Size Firms	423	0.10	0.75
Large Size Firms	423	0.11	0.78
Test scale		**0.10**	**0.77**

to use but also consistent to incorporate in empirical analysis. All the alpha values exceed the threshold of 0.7, suggesting a heightened level of internal consistency and validity. The total scale value exceeds 0.7, so ensuring a substantial level of overall reliability.

Moreover, the correlation matrix and its coefficients can also be used as validation in empirical analysis. By examining the relationship between variables, the validity of constructs can be evaluated. Furthermore, the correlation matrix also helps to understand the extent to which the measures accurately reflect differences between different constructs, thus significantly contributing to the overall validity assessment. Another reason to use correlation analysis is to ensure that the variables used in this study are free from any multicollinearity issues. According to the correlation results reported in Table V, the constructs are valid and the variables are free from multicollinearity issues. These results validate the accuracy of the data and variables. Based on the reported results no correlation coefficient is beyond 0.8, thus the presence of autocorrelation is ruled out.

Multivariate regression is applied to explore the impact of AI on recruitment biases. The results revealed that AI has a positive relationship with all types of recruitment biases, e.g., conformity bias ($\beta = 0.004$, $t = 3.06$, $p < 0.01$), affinity bias ($\beta = 0.176$, $t = 4.46$, $p < 0.01$), halo bias ($\beta = 0.162$, $t = 4.70$, $p < 0.01$),

TABLE V
CORRELATION MATRIX

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
2	0.50^{a}													
3	0.08^{c}	0.45												
4	0.28^{a}	0.15	0.29											
5	0.27	0.41	0.69^{a}	0.32^{c}										
6	0.30^{c}	0.41^{a}	0.20	0.06	0.43^{a}									
7	0.20	0.08	-0.19^{a}	0.06^{a}	0.03^{c}	0.13								
8	0.13^{a}	0.25^{a}	−0.09	−0.22	0.09	0.65^{a}	0.26							
9	0.00^{b}	0.43	0.39^{a}	−0.32	0.29^{c}	0.33^{b}	−0.06	0.21^{b}						
10	0.18^{a}	0.00	0.13	0.10	0.09^{a}	0.01	0.19^{b}	0.00	0.13^{a}					
11	0.01^{c}	0.06^{a}	0.07	0.02	0.10^{a}	0.13^{b}	0.23	0.05^{a}	0.06^{a}	-0.49^{a}				
12	0.26^{a}	0.00	0.09^{c}	0.28^{c}	0.06^{a}	0.03	0.05^{a}	0.02^{b}	0.05	0.11^{b}	-0.16			
13	0.06^{a}	0.02	0.03^{a}	0.04^{c}	0.14	0.15^{a}	0.11^{a}	0.05	0.11^{a}	-0.20^{a}	0.60^{b}	-0.37^{c}		
14	0.24^{b}	0.16^{a}	0.04	0.08	0.13^{c}	0.25^{a}	0.11	0.28^{a}	0.21^{a}	0.00	0.02^{a}	-0.13^{a}	0.05^{a}	
15	0.14^{b}	0.07	0.04^{c}	0.24	0.07	0.06	0.06^{b}	0.06^{b}	0.09^{c}	0.04^{b}	−0.11	0.27^{a}	-0.10^{b}	-0.33^{a}

Note: a = p<0.01, b = p<0.05, c = p>0.1
1 = Use of AI Tools, 2 = Fairness in Use of AI Tools, 3 = Conformity Bias, 4 = Affinity Bias, 5 = Halo Bias, 6 = Recency Bias, 7 = Horn Bias, 8 = Contrast Bias, 9 = Recruitment Process Biasness, 10 = Foreign Owned Firms, 11 = Government Owned Firms, 12 = Partnership Firms, 13 = Company, 14 = Medium Size Firms, 15 = Large Size Firms

TABLE VI
REGRESSION ANALYSIS (MAIN RESULTS)

Multivariate Regression Models							
Unit of Observation – Cross Section							
Dependent Variable – Recruitment Biases							
	Conformity	**Affinity**	**Halo**	**Recency**	**Horn**	**Contrast**	**Recruitment Process**
Independent Variable							
Use of AI Tools	0.004^{***}	0.176^{***}	0.162^{***}	0.273^{***}	0.154^{***}	0.117^{**}	0.011^{**}
	(3.06)	(4.46)	(4.70)	(6.13)	(3.86)	(2.26)	(2.62)
Control Variable							
Foreign Owned Firms	0.292^{***}	0.195^{**}	0.195^{***}	0.001^{***}	0.111^{**}	0.109^{**}	0.465^{***}
	(4.26)	(2.05)	(2.35)	(3.01)	(2.16)	(1.90)	(3.81)
Government Owned Firms	-0.276^{***}	-0.263^{**}	-0.178^{*}	−0.162	-0.414^{***}	-0.133^{***}	-0.279^{*}
	(−3.30)	(−2.26)	(−1.75)	(−1.24)	(−3.53)	(−2.88)	(−1.87)
Partnership Firms	0.140^{*}	0.474^{***}	0.045	0.165	0.116	0.303^{**}	0.014^{***}
	(1.62)	(3.93)	(0.43)	(1.22)	(0.96)	(1.92)	(2.09)
Company	0.020^{**}	0.046^{***}	0.153^{*}	0.133^{***}	0.015^{***}	0.127^{***}	0.149^{***}
	(1.95)	(2.44)	(1.67)	(2.13)	(3.14)	(2.93)	(2.11)
Medium Size Firms	0.034^{***}	0.111	0.124	0.581^{***}	0.144	0.709^{***}	0.513^{***}
	(2.51)	(1.18)	(1.52)	(5.49)	(1.52)	(5.78)	(4.26)
Large Size Firms	0.033	0.395^{***}	0.053	0.360^{***}	0.003^{***}	0.072^{***}	0.030^{*}
	(2.47)	(4.07)	(3.62)	(3.29)	(4.03)	(3.57)	(1.74)
Constant	1.906^{***}	1.467^{***}	1.782^{***}	1.847^{***}	2.124^{***}	2.604^{***}	2.296^{***}
	(21.39)	(11.85)	(16.49)	(13.21)	(16.98)	(16.08)	(14.45)
Model Summary							
Number of Observations	423	423	423	423	423	423	423
F-Stat	4.06	13.81	8.46	14.84	7.98	7.31	6.30
Prob. > F	0.000	0.000	0.000	0.000	0.000	0.000	0.000
R^2	0.055	0.164	0.108	0.174	0.102	0.094	0.082
Adjusted R^2	0.041	0.152	0.095	0.163	0.089	0.081	0.069

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$, t statistics in parentheses

recency bias ($\beta = 0.273$, $t = 6.13$, $p < 0.01$), horn bias ($\beta = 0.154$, $t = 3.86$, $p < 0.01$), contrast bias ($\beta = 0.117$, $t = 2.26$, $p < 0.01$), and recruitment process biases ($\beta = 0.011$, $t = 2.62$, $p < 0.05$). So, these results prove that AI tools can significantly reduce recruitment biases, as given in Table VI.

All the regression models have statistically significant F-statistics values, indicating that they are neither mis-specified nor provide misleading or weak results. As far as the empirical models' R-squares and adjusted R-squares are concerned, they show low statistical values. In cross-sectional survey datasets

TABLE VII
REGRESSION ANALYSIS (ROBUST CHECK)

Multivariate Regression Models							
Unit of Observation – Cross Section							
Dependent Variable – Recruitment Biases							
	Conformity	**Affinity**	**Halo**	**Recency**	**Horn**	**Contrast**	**Recruitment Process**
Independent Variable							
Fairness in Use of AI Tools	0.288***	0.127***	0.027*	0.400***	0.069*	0.260***	0.545***
	(11.39)	(3.18)	(1.76)	(9.37)	(1.72)	(5.14)	(12.26)
Control Variables							
Foreign Owned Firms	0.267***	0.263***	0.302***	0.086	0.175*	0.081***	0.419***
	(4.46)	(2.79)	(7.96)	(0.85)	(1.83)	(2.67)	(3.99)
Government Owned Firms	−0.212***	−0.255**	−0.012	−0.104***	−0.412***	−0.178***	−0.158***
	(−2.84)	(−2.17)	(−0.15)	(−2.83)	(−3.46)	(−2.20)	(−3.21)
Partnership Firms	0.168**	0.629***	0.210**	0.093	0.016	0.184***	0.042***
	(2.27)	(5.37)	(2.08)	(0.74)	(0.13)	(2.24)	(3.32)
Company	0.002***	0.031***	0.268***	0.268***	0.079***	0.064***	0.191*
	(3.03)	(2.27)	(3.11)	(2.40)	(2.75)	(2.48)	(1.64)
Medium Size Firms	0.063	0.065	0.088	0.583***	0.198**	0.679***	0.695***
	(1.07)	(0.69)	(1.14)	(5.84)	(2.10)	(5.73)	(6.68)
Large Size Firms	0.012***	0.389***	0.026***	0.386***	0.001***	0.089***	0.070***
	(2.19)	(3.96)	(2.32)	(3.68)	(2.01)	(2.72)	(3.64)
Constant	1.083***	1.519***	1.332***	1.342***	2.290***	2.130***	0.748***
	(12.10)	(10.76)	(11.26)	(8.89)	(16.01)	(11.89)	(4.76)
Model Summary							
Number of Observations	423	423	423	423	423	423	423
F-Stat	23.66	12.20	17.63	22.91	6.14	10.63	29.70
Prob. > F	0.000	0.000	0.000	0.000	0.000	0.000	0.000
R^2	0.252	0.148	0.201	0.246	0.080	0.131	0.297
Adjusted R^2	0.241	0.136	0.189	0.235	0.067	0.119	0.287

*$p<0.1$, **$p<0.05$, ***$p<0.01$, t statistics in parentheses

with 423 observations, it's common to observe a low value of R-Square. Several factors contribute to this phenomenon. First, numerous complex and often unobserved factors influence human behavior and are often the focus of surveys, making it challenging to fully explain the variables included in the model. Additionally, survey data frequently contain measurement errors, further reducing the ability of the independent variables to fully explain the variation in the dependent variable. Second, the population heterogeneity in cross-sectional data means that finding a single model that accurately predicts the dependent variable for all individuals is challenging.

Moreover, regarding the external validity of empirical results, we have changed the proxy of the independent variable from "Use of AI Tools" to "Fairness in Use of AI Tools." Such a change in the independent variable proxy not only validates the main regression results but also shows the robustness of our findings. Based on the results reported in Table VII, we not only testified to our main findings, but we also found supportive evidence regarding the selection of parametric analysis. The results mentioned in Table VII verify the findings and document that fair usage of AI tools subsequently reduces recruitment biases, e.g., conformity bias ($\beta = 0.288$, $t = 11.39$, $p < 0.01$), affinity bias ($\beta = 0.127$, $t = 3.18$, $p < 0.01$), halo bias ($\beta = 0.027$, $t = 1.76$, $p < 0.10$), recency bias ($\beta = 0.400$, $t = 9.37$, $p < 0.01$), horn bias ($\beta = 0.069$, $t = 1.72$, $p < 0.10$), contrast bias ($\beta = 0.260$, $t = 5.14$, $p < 0.01$), and recruitment process biases ($\beta = 0.545$, $t = 12.26$, $p < 0.01$). All of the models presented a statistically significant value of F-statistics and an acceptable level of R-square and adjusted R-square.

The results regarding the control variables suggested that, as compared to domestic-owned firms, employees faced less recruitment bias in foreign-owned firms. On the contrary, in the case of government-owned firms, recruitment biases are higher. Similarly, as compared to sole proprietorship firms, employees faced less recruitment bias in both partnerships and companies. Finally, as compared to small firms, employees encounter less recruitment bias in medium and large firms.

V. Discussion

By reducing biases that historically hamper recruiting decisions, AI has the potential to make a big impact on the recruitment process [65]. Although AI is not a magic bullet, it has many benefits that make hiring fairer and more objective [50]. AI's capacity for blind screening is a critical factor. Reviewing resumes and applications that include personal identifiers like names, photographs, or addresses is a common step in the traditional hiring process [35]. These identifiers may cause recruiters to have unconscious prejudices, potentially disqualifying qualified applicants. AI technologies can remove this information,

TABLE VIII
SUMMARY OF FINDINGS

Recruitment Biases	Hypotheses	AI Tools	
		Usage	Fairness
Conformity Bias	H_1	Proved	Proved
Affinity Bias	H_2	Proved	Proved
Halo Bias	H_3	Proved	Proved
Recency Bias	H_4	Proved	Proved
Horn Bias	H_5	Proved	Proved
Contrast Bias	H_6	Proved	Proved
Overall Recruitment Process Biasness	H_7	Proved	Proved

ensuring that the initial screening solely focuses on credentials and abilities. As a result, there may be less discrimination against candidates based on their gender, ethnicity, or socioeconomic position [66], [67].

The ability of AI to reduce numerous human biases is one of its key benefits. Some of the prejudices that frequently seep into the recruiting process are confirmation bias, affinity bias, and cultural bias [68]. AI systems, appropriately created and trained, can reduce these biases by focusing on the data and predetermined criteria [35]. As a result, AI systems avoid biases that can skew human judgment, evaluating candidates solely on their qualifications and skills. AI tools also enable data-driven decisions. They can find patterns of successful hires based on objective criteria by analyzing enormous databases [69]. Using a data-driven approach allows recruiting decisions to be based on facts rather than emotions, potentially reducing biases that could otherwise influence human decision-making.

We clarify the complex link between AI use and hiring biases within firms, thereby adding to both technological and socio-technical theory. The socio-technical approach reveals that social structures and organizational procedures, in addition to technical characteristics, influence technological developments like AI. We find that the application of AI technologies has a substantial impact on several recruitment biases. We specifically discovered a link between increased use of AI technologies and a reduction in conformity bias, affinity bias, halo bias, recency bias, horn bias, and contrast bias. This confirms the socio-technical theory, according to which AI is a technology that is ingrained in organizational procedures and social structures, influencing and being impacted by human prejudices and behaviors.

As compared to the previous studies, our results (see Table VIII) add to the larger body of research on organizational behavior and technology adoption. We further our knowledge of how technology interacts with human decision-making processes within organizational contexts by looking at the effect of AI usage on recruiting biases. We find our empirical findings intriguing and they reinforce the narrative we presented in the hypotheses section. Moreover, the results emphasize the need for companies to critically assess the use of AI technologies to reduce bias in the hiring process. Finally, our assessments validate the robustness of our findings, thereby enhancing their validity and dependability.

VI. Conclusion

From personal assistants like Siri and Alexa to self-driving cars, medical diagnosis, and financial analysis, AI has already started to change our lives in a variety of ways [70]. Overall, as new applications and technologies emerge, the relationship between AI and life remains complex and constantly evolving. It is crucial to continuously monitor and assess how AI affects our daily lives to ensure its use benefits humanity and safeguards our values and interests [71]. The field of AI, which is currently in a state of rapid development, focuses on developing devices and computer programmers that are capable of carrying out operations that typically require human intelligence, such as perception, reasoning, learning, and decision-making [72]. HR departments can use AI to improve decision-making, streamline processes, and improve employee experiences [73]. HR professionals can automate some tasks, learn about workforce trends, and make better talent management decisions by utilizing AI tools and technologies.

Keeping in view the said narrative, we have gathered data from the workers employed in the manufacturing sector in China. We found that using AI tools in the hiring process can significantly reduce recruitment biases using statistical evaluation techniques. Confirmation bias, recency bias, halo effect, horn effect, affinity bias, and contrast bias are some of the biases we have discussed in this study, and the findings support our hypothesis and are in line with the theoretical arguments built in this study. The findings suggested that AI solutions may benefit HR departments in several ways, including chatbots, predictive analytics, performance management, employee engagement, and recruiting and selection.

AI tools could improve hiring efficiency, objectivity, and consistency, thus introducing important policy implications in this context. Policymakers must create a strong regulatory framework to assure fairness and an unbiased, AI-driven recruitment system. This paradigm should be adaptable to the quickly changing AI ecosystem and emphasize fairness, openness, and responsibility. Policymakers can encourage organizations to use AI ethically in recruitment by setting explicit principles and criteria. We also need policy measures like comprehensive algorithmic auditing. Regular audits of recruitment AI algorithms are necessary to identify and rectify any biases. Independent auditors can improve transparency and trust, giving job seekers confidence that their applications are fair. Audit procedures, as part of policy, can help to ensure accountability and reduce bias in AI-driven recruitment systems.

Policy matters should also prioritize education and training. Policymakers shall support education for HR professionals, recruiting managers, and AI developers. These programs should help professionals understand AI bias, its effects, and how to mitigate it. By investing in education, policymakers can help workers make educated AI recruitment decisions. Data collection and openness in AI-driven recruitment should also be legislative priorities. Encouraging enterprises to disclose AI tool data and routinely check data sources for biases is critical. Strong policies should ensure data are representative and unbiased, fostering transparency and trust among job seekers using AI-driven recruitment platforms. Furthermore, we can integrate these AI systems with incentive systems to effectively reward employees who put in more effort and ultimately receive higher salaries, or vice versa.

Despite our study's insightful information about AI's potential to reduce hiring biases, it is important to consider some limitations. First, the specific sector in which we conducted our study, and the small sample of participants may restrict the generalizability of our findings to other contexts or populations. To assess the generalizability of our findings, future research should attempt to replicate our study using more diverse samples and in different industries. Second, our study relied on participant self-reported data, which could contain biases or inaccuracies. To provide a more thorough assessment of the effect of AI on reducing biases in recruitment, future research could include additional techniques like behavioral observation or objective performance measures. Finally, while our study focused on AI's potential to reduce hiring biases, it is crucial to recognize that improper development and application of the algorithm can lead to the emergence of new biases. Future studies should examine the potential for new biases to appear and devise methods to lessen them.

Future research on employing AI tools to reduce recruiting biases should explore novel approaches to make AI algorithms more transparent and interpretable, helping job seekers and hiring organizations comprehend the decision-making process. AI recruitment systems require more advanced continuous monitoring and real-time bias detection methods to develop adaptive models that can self-correct and adjust to shifting biases. Researchers should also investigate how AI might minimize prejudices and actively promote diverse and inclusive workplaces. We should prioritize standardized, independent auditing mechanisms for recruitment AI algorithms to evaluate fairness and accountability. We must explore AI in hiring within ethical and legal frameworks to comply with data privacy, anti-discrimination, and individual rights laws. Fair, honest, and ethical AI-driven recruitment practices will depend on these future research directions.

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